

Industrial Internship Report on “Prediction of Agriculture Crop Production in India”

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Executive Summary

This report provides details of the Industrial Internship facilitated by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). The internship provided an opportunity to study data science, statistics and machine learning concepts and apply them to a practical agriculture-focused problem.

The assigned project was “Prediction of Agriculture Crop Production in India”. The project involved preparing historical crop data and developing a supervised machine learning regression model to predict Production using Area and Yield as input features. The implementation was completed in Python using a Jupyter Notebook workflow.

The final dataset used for modeling contained 275 crop-year records covering 55 crops and five agricultural years from 2006–07 to 2010–11. Linear Regression and Random Forest Regression were evaluated. Random Forest performed better on the held-out test set with an MAE of 4.9701, RMSE of 8.7107 and R^2 score of 0.978039. The project gave practical exposure to data preprocessing, exploratory data analysis, regression, model evaluation and feature-importance analysis.

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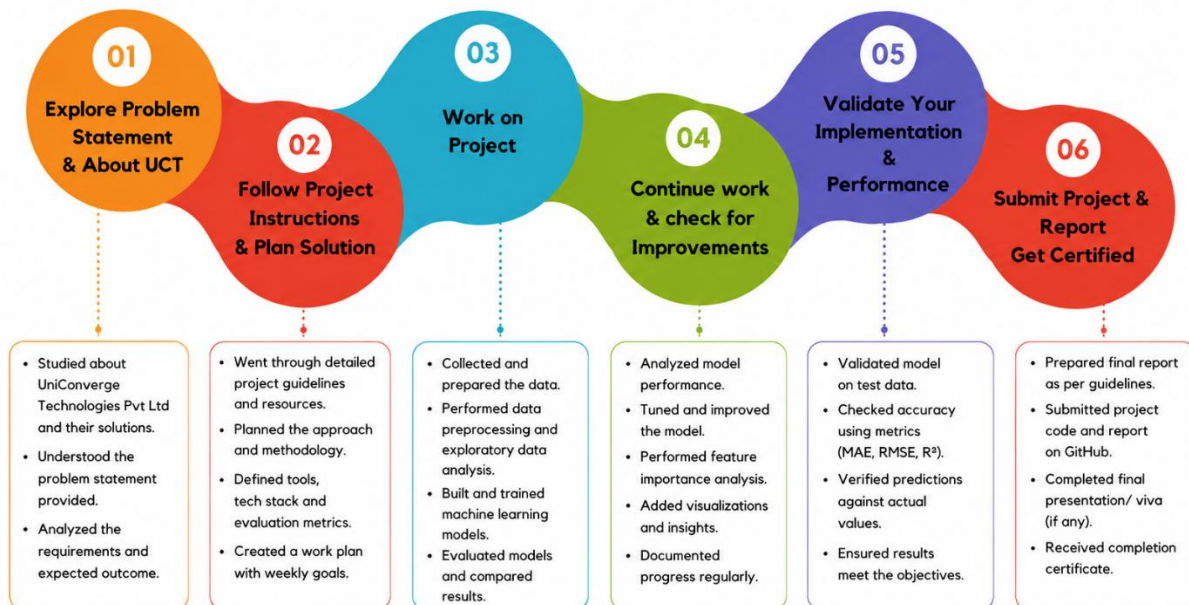
1 Preface

The six-week internship provided an opportunity to build practical knowledge in data science, probability and statistics, and machine learning. The learning phase covered introductory data science and machine learning concepts, followed by study of probability and statistics and machine learning concepts and tools. The final phase focused on applying these concepts to the assigned agriculture crop production prediction project.

The need for this internship was to bridge academic learning with practical implementation. Rather than studying algorithms only theoretically, the project required understanding a real dataset, preparing it for machine learning, training models, evaluating their performance and interpreting the results.

The assigned project, **Prediction of Agriculture Crop Production in India**, focuses on predicting crop Production from Area and Yield. The project was planned as a sequence of dataset understanding, preprocessing, exploratory analysis, model development, evaluation and final documentation.

The internship also provided exposure to an industry-oriented learning environment through upskill Campus, The IoT Academy and UniConverge Technologies Pvt Ltd. The overall experience improved my understanding of how data science and machine learning concepts can be applied to a practical problem.



I would like to thank the internship coordinators, mentors, faculty members and everyone involved in providing guidance and learning resources throughout the internship. I would also like to thank my college for supporting my participation in the internship.

To my juniors and peers, my message is to use internships as an opportunity to build practical skills, not only to complete assignments. Understanding the reason behind each step of a project is more valuable than simply obtaining the final output.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



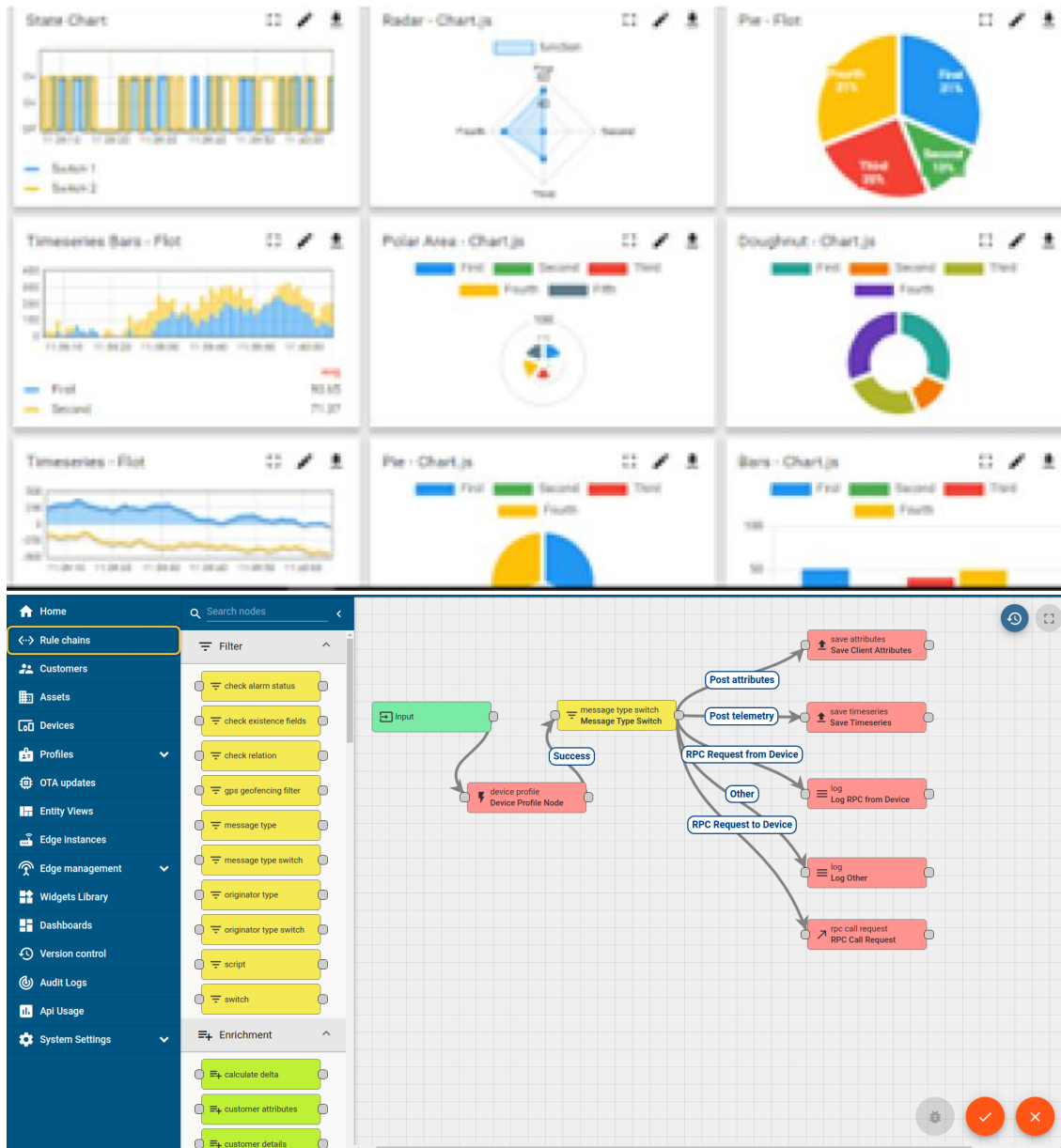
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



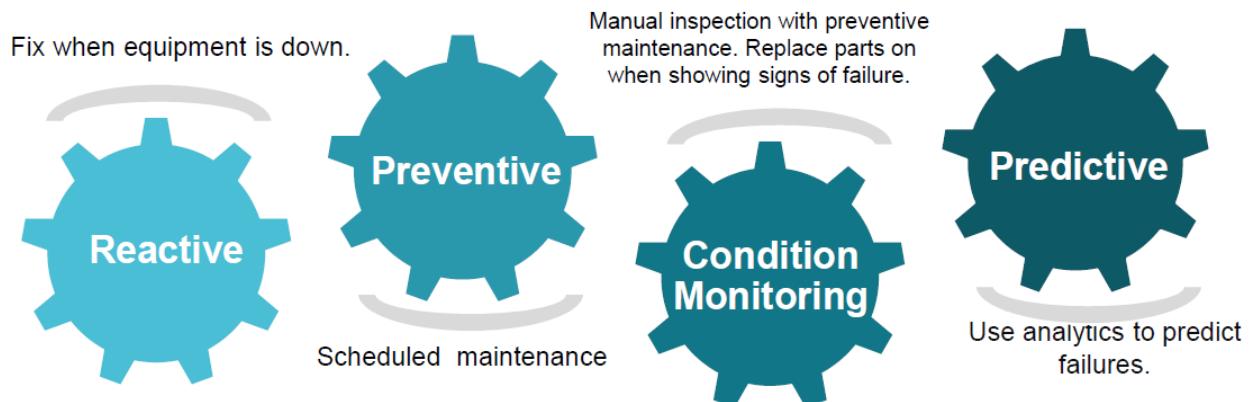


iii. based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

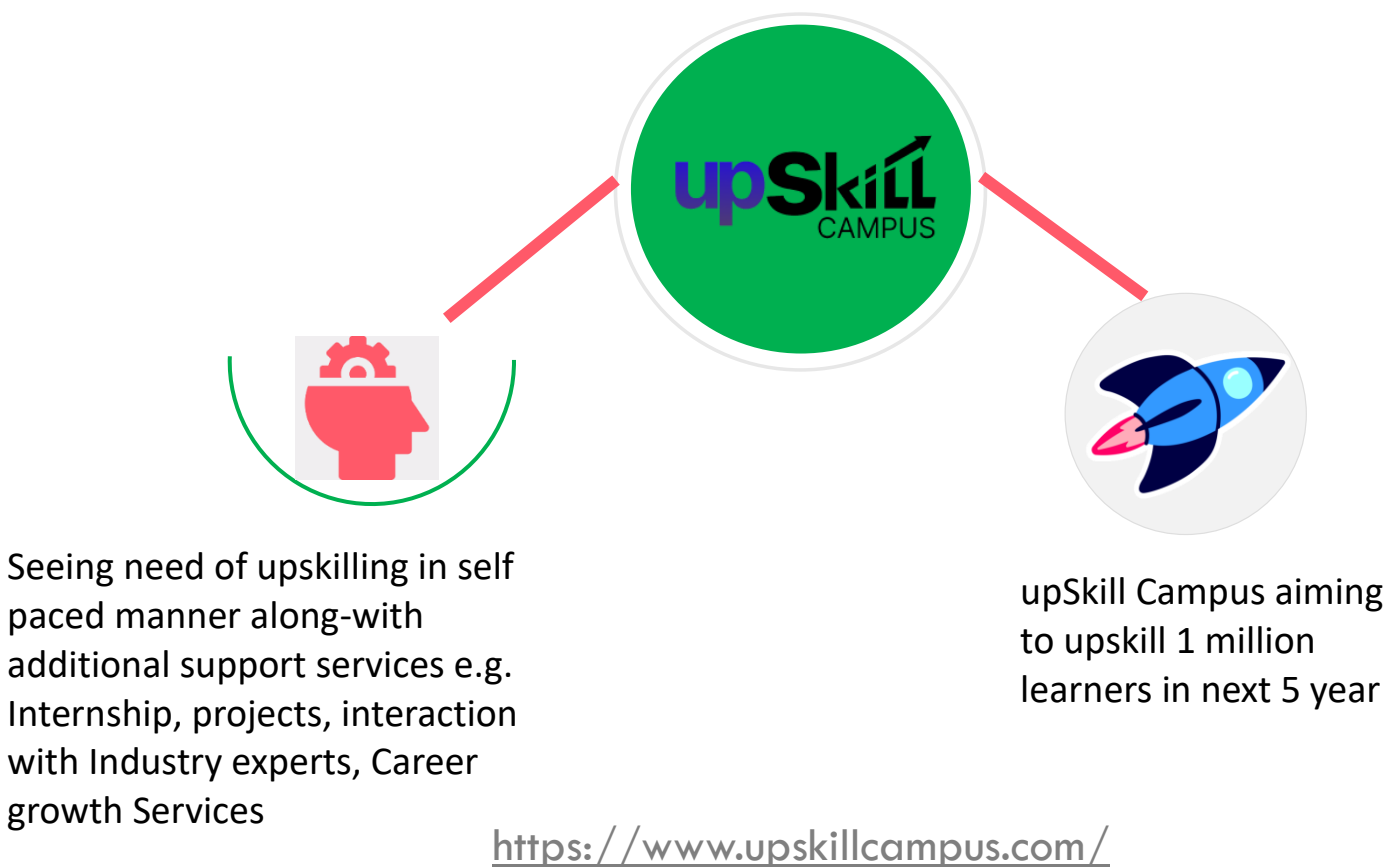
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

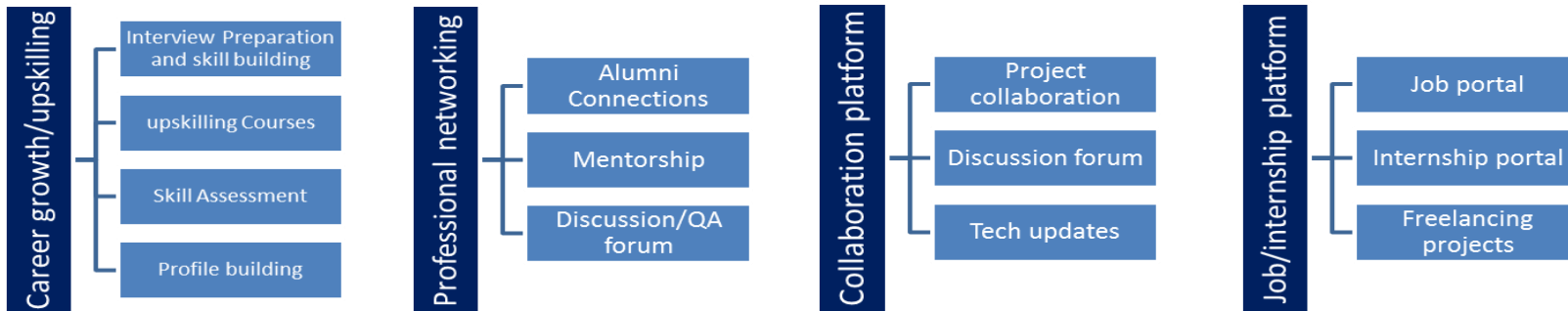


2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.





2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.
- ▣ to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Internship project description and dataset information supplied as part of the Agricultural Crop Production project.
- [2] Scikit-learn documentation used for machine learning algorithms and evaluation metrics.
- [3] Internship learning material provided through upskill Campus / The IOT Academy, including introductory Data Science and Machine Learning resources.

2.6 Glossary

Terms	Acronym
Artificial Intelligence	AI- field concerned with systems performing tasks requiring intelligence.
Machine Learning	ML- methods that learn patterns from data for prediction or decision making.
Data Science	DS – use of statistics, programming and analytical methods to obtain insights from data.
Linear regression	Regression algorithm modeling a linear relationship between features and target.
Random Forest	Ensemble machine learning algorithm using multiple decision trees.

3 Problem Statement

Agriculture is a major sector in India and crop production varies across crops and agricultural years. Historical agricultural data can be analyzed to understand production patterns and develop predictive models.

The assigned problem is to develop a machine learning regression solution for predicting agricultural crop Production in India using historical crop information. In the implemented solution, Area and Yield are used as input features and Production is used as the target variable.

The project dataset originally contained crop-wise production, area and yield values for five agricultural years. The data was transformed into 275 crop-year observations, analyzed and used to train regression models.

4 Existing and Proposed solution

Existing agricultural analysis can use historical production statistics, spreadsheets, reports and conventional statistical analysis. Such approaches are useful for understanding past data, but a predictive machine learning model can provide an automated way to estimate Production from selected input variables.

The proposed solution is a supervised regression pipeline. The historical dataset is cleaned and reshaped, exploratory analysis is performed, Area and Yield are selected as features, and regression models are trained to predict Production.

Two models were evaluated: Linear Regression as a baseline and Random Forest Regression as an ensemble model. The model with the stronger test-set performance was selected as the final model.

Value addition of the proposed solution:

1. Converts the supplied agricultural data in machine learning ready structure.
2. Provides automated production prediction from Area and Yield.
3. Compares more than one regression approach using objectives metrics.
4. Provides feature-importance analysis to understand the contribution of the inputs.
5. Saves the final trained for future inference.

4.1 Code submission (Github link)

Github link: <https://github.com/dighadepranav/upskillcampus>

4.2 Report submission (Github link) :

Report link: <https://github.com/dighadepranav/upskillcampus/blob/main/REPORT.md>

5 Proposed Design/ Model

The proposed model follows a sequential data science and machine learning workflow. The original wide-format dataset is first inspected and transformed into crop-year records. Exploratory analysis is then used to understand the numerical variables and their relationships. Area and Yield are selected as features, Production is selected as the target, and the records are split into training and testing sets. Linear Regression and Random Forest models are trained and evaluated. Random Forest is retained as the final model because it provides the better test-set performance.

5.1 High Level Diagram (if applicable)

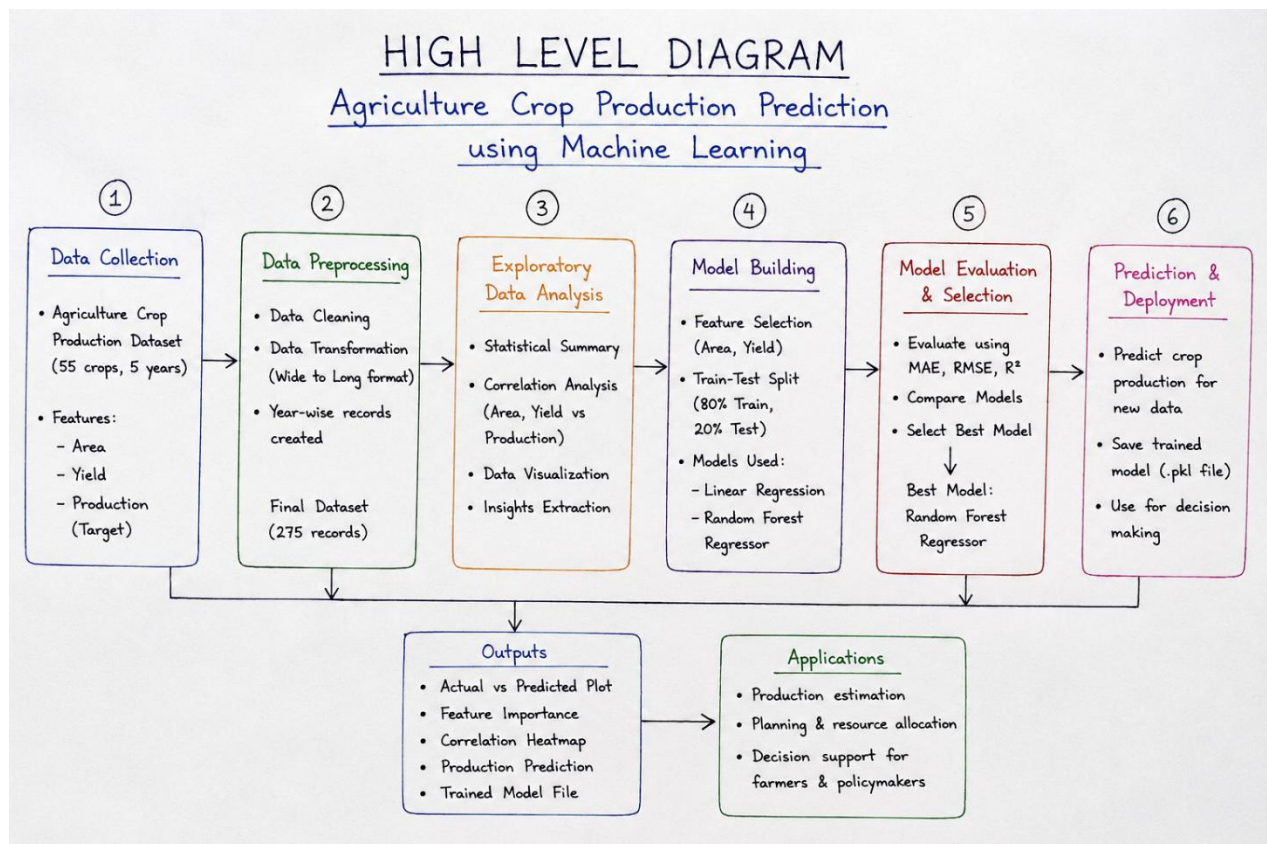
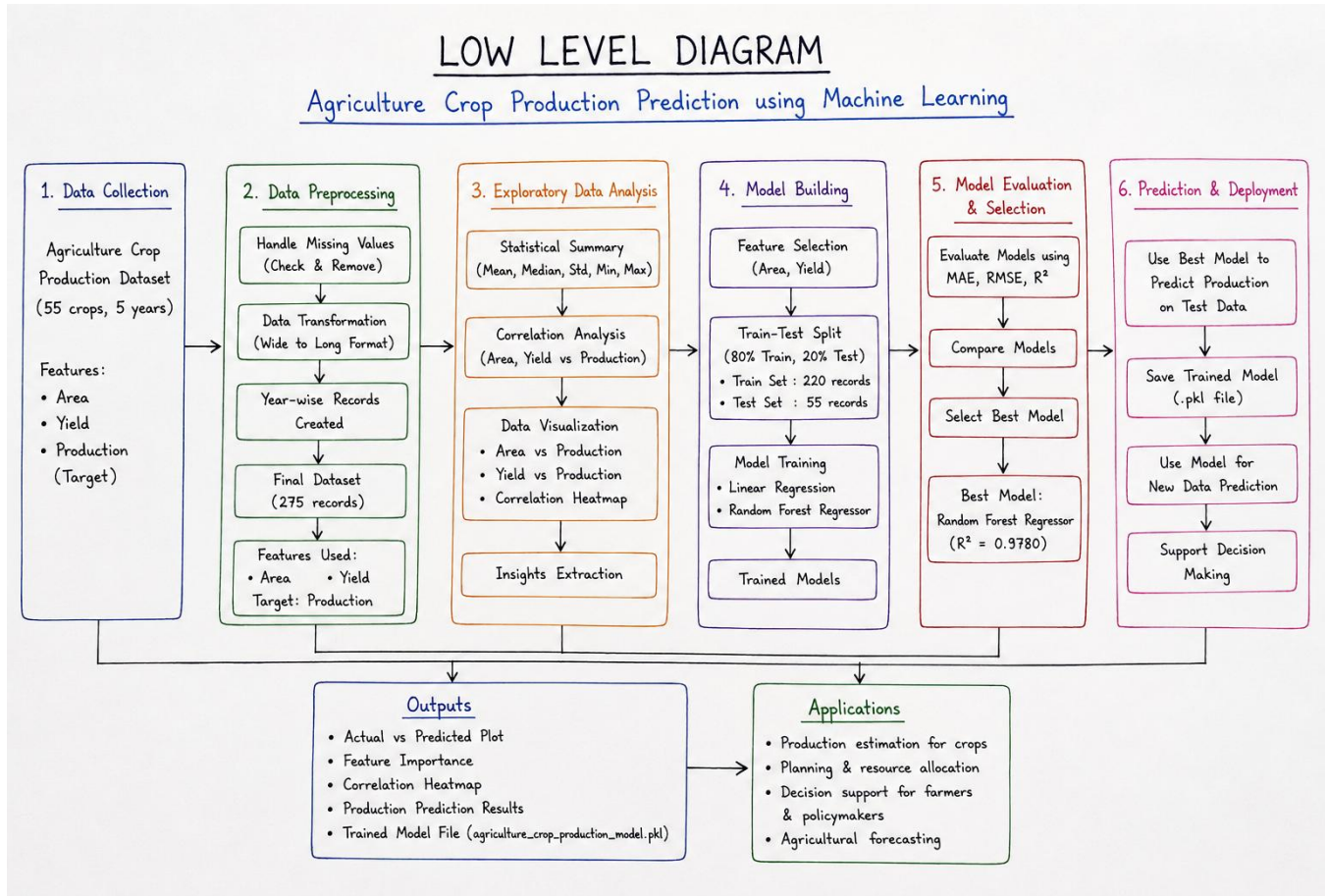


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

The implemented project is a notebook-based machine learning solution and does not contain a separate graphical user interface or production web interface. Therefore, a GUI interface is not applicable to the current scope.

The effective input interface of the model consists of two numerical values: Area and Yield. The model returns a numerical Production prediction. The project also includes the saved model file for programmatic inference.

6 Performance Test

Performance was evaluated as a supervised regression problem. The principal constraints considered were prediction error and explanatory performance. MAE, RMSE and R^2 were used to evaluate the model on data that was not used during training.

The available dataset is relatively small, with 275 crop-year records. Therefore, the results should be treated as performance on the available test split rather than a guarantee of future agricultural prediction accuracy.

6.1 Test Plan/ Test Cases

Test Case	Purpose	Expected Result
TC01 – Dataset Validation	Check structure, missing values and duplicates.	Dataset loads successfully; no missing or duplicate records in working data.
TC02 – Data transformation	Convert year-wise columns to crop-year format.	275 structured crop-year records are produced
TC03 – Feature/target preparation	Prepare Area and Yield as features and Production as target	Feature matrix shape = (275, 2); target shape = (275,).
TC04 – Train/test split	Separate training and testing records.	220 training and 55 testing samples.
TC05 – Linear Regression	Train baseline regression model.	Model produces test predictions and evaluation metrics.
TC06 – Random Forest	Train ensemble regression model.	Model produces test predictions and evaluation metrics.
TC07 – Model comparison	Compare MAE, RMSE and R^2 .	Random Forest performs better than Linear Regression.

6.2 Test Procedure

- Load and inspect the dataset.
- Transform the original year-specific columns into the modeling structure.
- Check missing values and duplicate records.
- Perform exploratory and correlation analysis.
- Select Area and Yield as features and Production as target.
- Split the data into 80% training and 20% testing sets.
- Train Linear Regression and Random Forest models using the training data.
- Predict Production for the test set.
- Calculate MAE, RMSE and R^2 .
- Compare the models and select the better-performing model.
- Generate actual-versus-predicted and feature-importance visualizations.
- Save the final Random Forest model using Joblib.

6.3 Performance Outcome

Model	MAE	RMSE	R^2 Score
Linear Regression	7.855755	11.724542	0.960213
Random Forest	4.970052	8.710653	0.978039

Random Forest is the selected final model because it has the lowest MAE and RMSE and the highest R^2 score. Its test-set R^2 of 0.978039 indicates that approximately 97.8% of the observed variation in the test-set Production values was explained by the model. This is a regression performance measure and should not be reported as 97.8% classification accuracy.

Feature Importance Outcome:

Feature	Importance
Yield	0.90573/90.57%
Area	0.09427/9.43%

The model's feature-importance analysis identifies Yield as the dominant input feature in the trained Random Forest

7 My learnings

The internship helped me move from theoretical understanding toward practical implementation. During the learning phase, I studied introductory Data Science and Machine Learning concepts, Python tools used for data analysis, probability and statistics concepts, and machine learning algorithms and evaluation methods. I also completed the internship quiz with a score of 100%.

The main technical learnings from the project were:

1. Understanding how a real dataset is inspected before modeling.
2. Using Pandas and NumPy for data preparation and numerical analysis.
3. Reshaping wide-format data into a year-wise modeling dataset.
4. Performing exploratory data analysis and interpreting statistical summaries.
5. Using correlation analysis to study relationships between variables.
6. Formulating a supervised regression problem with features and a target.
7. Splitting data into training and testing sets.
8. Training and comparing Linear Regression and Random Forest models.
9. Interpreting MAE, RMSE and R^2 correctly.
10. Understanding Random Forest feature importance.
11. Visualizing actual versus predicted values.
12. Saving a trained machine learning model with Joblib.
13. Improving problem-solving skills by debugging notebook errors and resolving Python environment issues.

These skills will support my future career in software, data and machine learning-related roles by improving my ability to understand datasets, implement algorithms and evaluate solutions systematically.

8 Future work scope

1. Expand the dataset to cover more historical years and more detailed state, district or regional information.
2. Include additional predictive variables such as rainfall, temperature, soil properties, irrigation, fertilizer use and pest information.
3. Evaluate additional machine learning algorithms such as Gradient Boosting, XGBoost and other ensemble methods.
4. Perform cross-validation and hyperparameter tuning to obtain more robust estimates of model performance.
5. Use time-aware validation and future-period testing when sufficient chronological data is available.
6. Develop a web or mobile application where users can enter Area and Yield and obtain a Production estimate.
7. Build dashboards for crop-wise, year-wise and region-wise agricultural analysis.
8. Monitor model performance over time and retrain the model when new agricultural data becomes available.